

7 (a) (i) 1.	$V_{rms} = \frac{V_0}{\sqrt{2}}$ $V_0 = 17\sqrt{2} = 24 \text{ V}$
2.	$\omega = 2\pi f$ $f = \frac{\omega}{2\pi} = \frac{380}{2\pi} = 60.5 \text{ Hz}$
(a) (ii)	The power dissipated in the heat coil depends on the rms current and not the average current. The rms current produces the same power in an AC circuit as the direct current with the same value in a DC circuit. The alternating current travels back and forth in opposite directions and cause heating in the

	coil.
(b) (i)	The specific heat capacity c of a substance is defined as the heat (thermal energy) per unit mass required to raise the temperature of the substance by one unit of temperature.
(ii)	The intercept represents a systematic error which is the rate of heat loss from the apparatus to the environment.
(iii)	$Pt = mtc\Delta\theta + H_{loss}$ $P = mc(3.8) + \frac{H_{loss}}{t}$ $m = \frac{1}{3.8c}P - \frac{H_{loss}}{3.8ct}$ From the graph, $\text{gradient} = \frac{3.8 - 1.3}{75 - 35} = 0.0625$ $0.0625 = \frac{1}{3.8c}$ $c = 4.21 \text{ J g}^{-1} \text{ K}^{-1} = 4210 \text{ J kg}^{-1} \text{ K}^{-1}$
(c)	From the equation of the graph above, the gradient of the new graph is the same as specific heat capacity and temperature difference remains unchanged. As the temperature of the apparatus is higher, the rate of heat loss is higher hence the y-intercept is more negative.
(d) (i)	The first law of thermodynamics state that the increase in internal energy of a system is equal to the sum of the heat supplied to the system and work done on the system.
(ii)	At constant volume, there is no work done by the system. Hence according to the first law of thermodynamics, the heat supplied is only used to increase the internal energy. At constant pressure, heat is supplied to increase the internal energy and for the gas to do work against the environment. Hence, the constant pressure specific heat capacity is higher than that of constant volume. <i>Note: Most were able to state the effect of constant pressure on work done but there was confusion as to whether the work would be done on the gas or by the gas.</i>
8 (a) (i)	Simple harmonic motion is an oscillatory motion where the acceleration of the